CALIBRATED FORECASTING AND PERSUASION

ATULYA JAIN AND VIANNEY PERCHET

Abstract. We study a dynamic game where an expert sends probabilistic forecasts to a decision-maker. The decision-maker verifies these forecasts using a calibration test based on past data. How should the expert send forecasts to maximize her payoff while passing the test? For a stationary ergodic process, we characterize the optimal forecasting strategy by reducing the dynamic game to a static persuasion problem. Calibration restricts long-run forecast distributions to mean-preserving contractions of the distribution of truthful forecasts. We compare the payoffs attainable by an informed and uninformed expert, providing a benchmark for the value of information. Finally, we consider a decision-maker who uses a mean-based learning strategy and show that the expert can guarantee at least the calibration benchmark and sometimes strictly more.

1. Introduction

Probability forecasts are widely used by experts to provide information about uncertain outcomes. These forecasts shape the beliefs of decision-makers and guide them to take specific actions. For example, an investor relies on forecasts by a financial analyst to determine which asset will perform best. However, the decision-maker follows the expert's forecasts only if they are credible. A standard way to determine credibility is to perform statistical tests on the realized outcomes and verify the expert's claims. We focus on an objective test: *calibration*, which is based on the frequentist interpretation of probability. The idea is to check whether the forecast of an outcome is close to the actual proportion of times the outcome occurred when the forecast was made. For example, the investor checks whether an asset outperformed others on 70% of the days when the analyst predicted a 0.7 chance of it being the best. Note that a single day's outcome–whether the asset actually outperformed others–cannot prove the expert right or wrong. However, over time, one can collect data on the forecasts and the realized outcomes to evaluate the

Date: October 29, 2026.
Department of Economics, University of Bonn. Email: ajain@uni-bonn.de.
ENSAE and Criteo AI Lab. Email: vianney.perchet@normalesup.org.
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expert's credibility. Calibration is central to forecasting and is used to assess the accuracy of prediction markets [Page and Clemen, 2012].

The decision-maker relies on accurate forecasts to take optimal actions. But in many settings, the expert herself has skin in the game and is affected by the decision-maker's action. This misalignment of preferences leads to *strategic forecasting*. For instance, if a financial analyst earns a substantial commission upon the purchase of a certain asset, her forecasts may be biased in favor of this asset. We study the extent of an expert's payoff gain from strategic forecasting in an infinite horizon dynamic game. Our main focus is on an *informed expert* who knows the data-generating process and can produce calibrated forecasts by providing truthful forecasts. Given that she is tested by a calibration test, how should the expert send forecasts to maximize her payoff?

To do this, we develop a dynamic sender-receiver model where the state of the world evolves according to a stochastic process. In each period, the sender first provides a probabilistic forecast of that period's state to the receiver. The receiver then uses a calibration test to assess the sender's credibility. If the sender passes the test, the receiver takes the forecasts at face value and chooses a best response to this forecast. Otherwise, the sender incurs a punishment cost in that period. We focus on high punishment costs, since otherwise the sender may prefer uncalibrated strategies. Finally, the realized state is observed before moving to the next period.

Our approach differs from standard Bayesian models, where the receiver is assumed to know the distribution of the process or at least to hold a prior over it. In such models, given the sender's strategy, the receiver updates beliefs from the observed forecasts and chooses actions accordingly. Handling such beliefs in equilibrium—even in simple environments—is complex and imposes strong rationality assumptions. In contrast, we assume the receiver has no prior information about either the process or the sender's strategy. Instead, he evaluates the sender's credibility using a calibration test on the history of forecasts and states.

We first analyze the case of an informed sender who knows the data-generating process. She can pass the calibration test by forecasting truthfully. However, other forecasting strategies may also pass the test and result in higher payoffs. We show that calibration restricts the feasible long-run distribution of forecasts. For a stationary and ergodic process, a finitely supported distribution of forecasts can arise under calibration if and only if it is a mean-preserving contraction of the distribution of truthful forecasts (Lemma 1). Intuitively, this corresponds to merging low-value and high-value truthful forecasts with appropriate weights so that the overall forecast matches the true frequencies. Thus, while the forecasts must be correct on average, they can be less informative than truthful forecasts. This characterizes the extent of strategic forecasting permitted by calibration.

Our main result (Theorem 1) establishes that the informed sender's maximum payoff in the dynamic forecasting game equals the value of a static persuasion problem. The state,

signal, and prior in the persuasion problem correspond to the conditional probability, the forecast, and the distribution of conditionals in the dynamic game. In Bayesian persuasion [Kamenica and Gentzkow, 2011], the sender commits to a signaling policy that maximizes her expected utility. We construct a forecasting strategy that implements an optimal signaling policy period by period, so forecasts depend only on the current conditional. For the upper bound, we show that any payoff gain from failing the calibration test is bounded proportionally to the frequency of failures and outweighed by sufficiently high punishment costs. Thus, the sender's dynamic problem reduces to a static persuasion problem. An interesting implication is that calibration can substitute for commitment: the sender achieves the Bayesian persuasion payoff. Guo and Shmaya [2021] obtain a related result in a static forecasting model with a miscalibration cost proportional to the expected distance between forecasts and their true conditional distributions. In our dynamic model, the receiver instead evaluates calibration using the history of forecasts and realized states, and the sender incurs a fixed cost whenever she fails the test.

We next consider an uninformed sender who does not know the data-generating process. As shown by Foster and Vohra [1998], even an uninformed sender can produce calibrated forecasts for any stochastic process. We derive payoff guarantees for the uninformed sender in two environments and relate them to benchmarks from Bayesian persuasion. First, in an adversarial setting where nature can adapt its strategy, there is no fixed prior. In this case, the sender can guarantee the closed convex envelope of her approximate indirect utility, evaluated at the empirical distribution of states (Theorem 2).[1] This parallels the sender's worst payoff in canonical Bayesian persuasion, with the empirical distribution of states playing the role of the prior. Second, when the unknown process is an ergodic Markov chain, the uninformed sender can do better. In particular, she can guarantee the finite-grid counterpart of the no-disclosure payoff in the persuasion problem with prior given by the distribution of conditionals—the same problem as in the informed sender case (Proposition 1). This comparison separates credibility from information. An expert need not know the data-generating process to produce calibrated forecasts; information is valuable because it allows the sender to benefit more from strategic forecasting.

We apply our model to analyze a financial platform that provides forecasts about a Markovian state. The platform's payoff depends on forecast precision and user engagement. This creates a trade-off: precise forecasts enhance reputation but reduce the time a user spends on the platform. We derive the platform's optimal forecasting strategy using the concavification approach and illustrate it graphically. We show that it may be optimal for the platform to send coarse forecasts rather than truthful ones.

One might question whether calibration is an effective decision-making heuristic, given that it permits strategic forecasting. To address this concern, we also model the receiver's behavior using regret minimization, a widely studied approach in online settings. Regret

[1]For the uninformed sender, we restrict forecasts to a finite set and focus on ϵ-calibration. Thus, our payoff guarantees are approximate.

compares the receiver's realized payoff to the payoff he would have obtained by playing the best fixed action in hindsight. Regret minimization is closely linked to calibration (see Perchet, 2014), and we use this connection to compare strategic forecasting under the two decision rules. We show that if the receiver myopically best responds to the forecasts of any calibrated strategy, then he incurs no regret, thereby justifying the use of calibration as a decision-making heuristic (Proposition 2). We next focus on a natural class of learning strategies, called mean-based algorithms. These algorithms assign vanishing probability to actions whose cumulative payoff is substantially lower than that of another action. Against a mean-based receiver, the informed sender can guarantee the calibration benchmark, namely the persuasion value, and sometimes strictly more (Proposition 3 and Theorem 3).

The rest of the paper is organized as follows. In Section 2, we introduce the dynamic forecasting game and the calibration test. In Section 3, we present the persuasion problem, state our main results and provide an application. In Section 4, we consider a receiver who minimizes regret instead of using the calibration test. Section 5 discusses the related literature. In Section 6, we conclude and discuss future work. All omitted proofs are in Appendix A.

2. Dynamic forecasting game

We consider a dynamic game between a sender (she) and a receiver (he), where the state of the world evolves over time. In each period, the sender sends a forecast about that period's unknown state. The receiver observes the forecast and then chooses an action. The realized state is observed before proceeding to the next period.

Let Ω denote the finite set of states, $F \subseteq \Delta(\Omega)$ denote the set of feasible forecasts over the states, and A denote the finite set of actions.[2] Unless specified otherwise, all forecasts are feasible, i.e., $F = \Delta(\Omega)$. The state $(\omega_t)_{t\geq 1}$ evolves over time and is governed by a stochastic process with distribution $\mu \in \Delta(\Omega^\infty)$. Denote by ω_t the state in period t, and by $\omega^t = (\omega_1, ..., \omega_{t-1})$ the history of states up to period t. We assume the sender is informed and knows the distribution μ, whereas the receiver is uninformed.[3] In particular, given the history ω^t, the sender can compute the *conditional probability* in period t, denoted by $p_t = \mu(\cdot \mid \omega^t) \in \Delta(\Omega)$. Hence, the sender knows the objective probability over states at each period. We assume that the conditionals take values from a finite set $D \subset \Delta(\Omega)$.[4]

In each period, the sender's forecast may depend on the entire history of past forecasts and states. Formally, a forecasting strategy is a map $\sigma : \bigcup_{t\geq 1}(F \times \Omega)^{t-1} \to \Delta(F)$. Let f_t denote the realized forecast in period t. After observing f_t, the receiver chooses an

[2] $\Delta(X)$ denotes the set of all probability distributions over the set X.
[3] Section 3.3 considers the case where the sender is uninformed.
[4] This is a restriction on the stochastic process μ, rather than on the set of feasible forecasts. It holds, for example, for finite-state Markov chains. The assumption is imposed for tractability; more general processes could be treated using finite approximations of the set of conditionals.

action $a_t \in A$ and then the state $\omega_t \in \Omega$ is realized. The sender's and receiver's payoffs depend only on the realized state and action and are given by $u_S(\omega_t, a_t)$ and $u_R(\omega_t, a_t)$, respectively. A forecasting strategy is *truthful* if $f_t = p_t$ in every period.

Calibration Test: To model the receiver's behavior, we adopt a frequentist approach based on calibration. The receiver has no prior information about the stochastic process or the sender's strategy. Instead, he evaluates the sender's credibility by comparing her forecasts with the realized frequencies of states. In each period, he computes the calibration error using the history of forecasts and states. The sequence of error margins $(\epsilon_t)_{t\geq 1}$ is fixed ex ante, with ϵ_t specifying the tolerance for calibration error over the first t periods.

Definition 1. A T-sequence of forecasts $(f_t)_{t=1}^T$ is ϵ_T-*calibrated* if

$$\sum_{f\in F} \frac{N_T[f]}{T} \|\overline{\omega}_T[f] - f\| \leq \epsilon_T, \tag{2.1}$$

where $N_T[f]$ and $\overline{\omega}_T[f]$ denote, respectively, the number of periods in which the forecast equals f up to period T and the empirical distribution of states in those periods:

$$N_T[f] := \sum_{t=1}^{T} \mathbf{1}_{\{f_t=f\}}, \qquad \overline{\omega}_T[f] := \frac{1}{N_T[f]} \sum_{t=1}^{T} \mathbf{1}_{\{f_t=f\}} \delta_{\omega_t},$$

where $\delta_\omega \in \Delta(\Omega)$ is the Dirac distribution on state ω and $\|\cdot\|$ is the Euclidean norm.[5]

A T-sequence of forecasts is ϵ_T-calibrated if the frequency-weighted average distance between forecasts and the empirical distributions of states does not exceed the error margin ϵ_T.[6] We illustrate the calibration test using a rain forecasting example.

Example 1. *Consider the states $\Omega = \{0, 1\}$, where $\omega = 1$ denotes rain and $\omega = 0$ denotes no rain. The weather evolves according to a Markov chain: with probability 0.8 the state remains the same from one period to the next, that is, $\mu(\omega_{t+1} = 1 \mid \omega_t = 1) = \mu(\omega_{t+1} = 0 \mid \omega_t = 0) = 0.8$ for every $t \in \mathbb{N}$. In each period, the sender predicts the probability of rain. Consider three sequences of forecasts, denoted $F1$, $F2$, and $F3$ (Table 1). $F1$ corresponds to truthful forecasts, where the sender announces 80% after rain and 20% after no rain. $F2$ corresponds to coarse forecasts obtained by garbling the truthful forecasts. $F3$ corresponds to extreme forecasts, predicting 100% after rain and 0% after no rain.*

For the error margin $\epsilon_{10} = 0.1$, both $F1$ and $F2$ pass the ϵ_{10}-calibration test, while $F3$ does not. For instance, when $F2$ predicts a 60% chance of rain, it rains in three out of five periods, and similarly for 40%. In contrast, when $F3$ predicts a 100% chance of rain,

[5] For completeness, if $N_T[f] = 0$, define $\overline{\omega}_T[f] = f$.

[6] This holds even if the set F is infinite, as the sum in the definition is implicitly restricted to the forecasts actually sent, which are finite.

TABLE 1. Sequences of forecasts and the calibration test.

Period	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	
State	*1*	*0*	*0*	*0*	*0*	*0*	*1*	*1*	*1*	*1*	
F1	*80%*	*80%*	*20%*	*20%*	*20%*	*20%*	*20%*	*80%*	*80%*	*80%*	✓
F2	*60%*	*60%*	*40%*	*40%*	*60%*	*40%*	*40%*	*60%*	*60%*	*40%*	✓
F3	*100%*	*100%*	*0%*	*0%*	*0%*	*0%*	*0%*	*100%*	*100%*	*100%*	✗

it does not always rain. Thus, both the truthful forecasts $F1$ and the coarse forecasts $F2$ pass the test. Intuitively, in the long run, calibration preserves the correct mean while allowing forecasts to be less informative than the truth.

Pass: The sender passes the calibration test in period t if the sequence of past forecasts $(f_i)_{i=1}^{t-1}$ is ϵ_{t-1}-calibrated.[7] In this case, the receiver takes the forecast at face value and chooses $a_t = \hat{a}(f_t)$, where $\hat{a}(f)$ denotes his best response to belief f, with ties broken in favor of the sender. Formally,

$$\hat{a}(f) \in \arg\max_{a \in A} \sum_{\omega \in \Omega} f(\omega) u_R(\omega, a).$$

Fail: If the sequence of past forecasts is not ϵ_{t-1}-calibrated, the sender fails the calibration test in period t. In this case, her payoff in that period is $-c$, where $c > 0$ is the punishment cost. The receiver refrains from playing according to the sender's forecast until she passes the test in a future period. This punishment can be interpreted as the receiver reverting to a default action.[8] Alternatively, it may reflect a loss of credibility, with an intermediary forwarding the sender's forecasts only when she passes the calibration test.

The sender's goal is to choose a forecasting strategy $\sigma^\star$ that maximizes her expected long-run average payoff:

$$\liminf_{T \to \infty} \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}_{\sigma^\star, \mu} \left[u_S(\omega_t, a_t) \right]. \tag{2.2}$$

We refer to the strategy $\sigma^\star$ as the *optimal forecasting strategy*. To make the problem tractable and economically meaningful, we impose two assumptions.

First, observe that for a sufficiently small error margin ϵ_T, even a truthful sender may fail the ϵ_T-calibration test with positive probability. For instance, in Example 1, when $\epsilon_1 = 0.1$, the truthful forecasts are not ϵ_1-calibrated regardless of which state ω_1 is realized. Nevertheless, as more data are observed, empirical frequencies converge to the truthful forecasts. This motivates an asymptotic notion of calibration.

[7] In period $t = 1$, there is no past data, so the sender passes the test trivially.

[8] This can be modeled with an additional default action a_d, used when the sender fails the calibration test, such that $u_S(\omega, a_d) = -c$ and $u_R(\omega, a_d) = 0$ for all $\omega \in \Omega$. In financial forecasting, for example, this may correspond to the investor not buying any asset, resulting in zero commission for the analyst.

Definition 2. A forecasting strategy σ is ϵ-*calibrated* if

$$\limsup_{T\to\infty} \sum_{f\in F} \frac{N_T[f]}{T} \|\overline{\omega}_T[f] - f\| \leq \epsilon, \quad \mathbb{P}_{\sigma,\mu}\text{-a.s.} \tag{2.3}$$

A forecasting strategy σ is *calibrated* if it is ϵ-calibrated, for every $\epsilon > 0$.

Under calibration, the empirical distribution of states conditional on any forecast used with positive long-run frequency converges to that forecast. The truthful forecasting strategy is calibrated for any stochastic process [Dawid, 1982].

A natural requirement is that the error margins in the finite calibration test converge to zero over time. However, if they shrink too quickly, even truthful forecasting may fail the test too often. We therefore impose the following requirement.

Assumption 1. *Fix a stationary ergodic process μ. The error margins satisfy $\epsilon_t \to 0$ and are such that, for every finite $F_0 \subseteq F$, any calibrated strategy induced by a fixed rule $\pi : D \to \Delta(F_0)$ fails the calibration test in a vanishing fraction of periods in expectation.*

Assumption 1 ensures that calibrated strategies induced by a fixed rule, including truthful forecasting, are not rejected too often. Proposition 4 in Appendix A shows that such a sequence exists for every stationary ergodic process; in fact, for this sequence, every such strategy then fails the test only finitely many times almost surely.[9]

Second, we use calibration as the credibility criterion for the sender. Although an informed sender can always be calibrated by forecasting truthfully, she may prefer an uncalibrated strategy when the punishment cost is low. We therefore focus on sufficiently high costs.

Assumption 2. *The punishment cost satisfies $c > (1+\sqrt{2})L$, where*

$$L := \max_{a\in A} \|u_S(\cdot, a)\|.$$

As we show later, sufficiently high punishment costs imply that any optimal forecasting strategy fails the calibration test in only a vanishing fraction of periods in expectation. This implication is not immediate: a sender may benefit from occasional failures if they make the distribution of states in the periods in which she passes more favorable. We show that any such gain is bounded proportionally to the frequency of failures.

3. Main results

Our main result characterizes the informed sender's optimal payoff for a stationary ergodic process by reducing the dynamic forecasting game to a static persuasion problem. Before stating the main result, we introduce the persuasion problem. We then analyze the payoff that an uninformed sender can guarantee. Finally, we illustrate the results with an application to a financial forecasting platform.

[9]Uniform rates of convergence need not hold across arbitrary strategies and processes unless stronger dependence conditions, such as mixing or irreducibility in Markov chains, are imposed.

3.1. **Persuasion problem**

We consider a static persuasion problem whose parameters are directly linked to those of the dynamic forecasting game. The state space $\Delta(\Omega)$ corresponds to conditional probabilities, and the prior $P \in \Delta(\Delta(\Omega))$ is given by their distribution. The sender commits to a signaling policy $\pi : \Delta(\Omega) \to \Delta(\Delta(\Omega))$, where signals correspond to forecasts.[10] Once the conditional $p \in \Delta(\Omega)$ is realized, the sender sends a forecast $f \in \Delta(\Omega)$ according to $\pi(\cdot \mid p)$. Each forecast induces a posterior belief over conditionals and, consequently, a posterior mean. Thus, a signaling policy induces a distribution of posterior means $Q \in \Delta(\Delta(\Omega))$. Without loss of generality, we restrict attention to policies under which the posterior mean induced by forecast f equals f. It is well known that a distribution Q is implementable by a signaling policy if and only if it is a mean-preserving contraction of the prior P (e.g., Gentzkow and Kamenica, 2016, Kolotilin, 2018, Kleiner et al., 2021, Arieli et al., 2023).

A probability distribution with finite support can be represented as $P = (\boldsymbol{\lambda}, \boldsymbol{p})$, where $\boldsymbol{p} = (p_1, \ldots, p_n)$, with $p_i \in \Delta(\Omega)$, denotes the support and $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_n)$ the associated weights. Similarly, write $Q = (\boldsymbol{\nu}, \boldsymbol{f})$, where $\boldsymbol{f} = (f_1, \ldots, f_m)$ denotes the support and $\boldsymbol{\nu} = (\nu_1, \ldots, \nu_m)$ the associated weights. We follow Elton and Hill [1992] and Whitmeyer and Whitmeyer [2021] in defining mean-preserving contractions.

Definition 3. A probability distribution $Q = (\boldsymbol{\nu}, \boldsymbol{f})$ is a *simple mean-preserving contraction* of $P = (\boldsymbol{\lambda}, \boldsymbol{p})$, if there exists a row-stochastic matrix $G \in \mathbb{R}^{n \times m}$ satisfying

$$\boldsymbol{\lambda} G = \boldsymbol{\nu}, \tag{3.1}$$

$$(\boldsymbol{\lambda}\boldsymbol{p}) G = (\boldsymbol{\nu}\boldsymbol{f}), \tag{3.2}$$

where $\boldsymbol{\lambda}\boldsymbol{p} = (\lambda_1 p_1, \ldots, \lambda_n p_n)$ and $\boldsymbol{\nu}\boldsymbol{f} = (\nu_1 f_1, \ldots, \nu_m f_m)$.

Intuitively, a simple mean-preserving contraction takes a fraction G_{ij} of the mass λ_i at each p_i and pools these to form mass ν_j at f_j.

Definition 4. Q is a *mean-preserving contraction* of P if there exists a sequence of simple mean-preserving contractions $(Q_m)_{m=1}^{\infty}$ such that $Q_m \to Q$ (weak convergence). Denote by $\mathcal{M}(P)$ the set of all mean-preserving contractions of P.

The sender's indirect utility from forecast f is given by

$$\hat{u}_S(f) := \sum_{\omega \in \Omega} f(\omega)\, u_S(\omega, \hat{a}(f)),$$

where $\hat{a}(f)$ is the receiver's best response to belief f.

Given prior P, the sender chooses $Q \in \mathcal{M}(P)$ to maximize her expected indirect utility. Define the *persuasion value* by

$$\mathrm{Per}(P) := \max_{Q \in \mathcal{M}(P)} \mathbb{E}_Q[\hat{u}_S]. \tag{3.3}$$

[10] To clarify, the sender does not have commitment power in the dynamic forecasting game.

Let $Q^\star$ denote an optimal distribution of forecasts solving (3.3), and let $\pi^\star$ denote a signaling policy that implements it.

When $\mathrm{Supp}(P)$ is affinely independent, the persuasion problem can be solved using the concavification approach [Aumann et al., 1995, Kamenica and Gentzkow, 2011] on the restricted domain $C = \mathrm{Co}(\mathrm{Supp}(P))$.[11] In particular,

$$\mathrm{Per}(P) = \mathrm{Cav}\, \hat{u}_S\big|_C(m(P)), \tag{3.4}$$

where $m(P) = \sum_{i=1}^n \lambda_i p_i$ denotes the mean of P and $\mathrm{Cav}\, \hat{u}_S\big|_C$ denotes the concave envelope of $\hat{u}_S$ on C.

This follows from a simple characterization of $\mathcal{M}(P)$. Under affine independence, feasibility reduces to Bayes plausibility on C: Q is feasible if and only if $\mathrm{Supp}(Q) \subseteq C$ and $m(Q) = m(P)$ [Wu, 2023]. The canonical Bayesian persuasion setting is the special case in which the sender is perfectly informed, so $\mathrm{Supp}(P) = \{\delta_\omega : \omega \in \Omega\}$.

By Bauer's maximum principle the maximum in (3.3) is attained at an extreme point of $\mathcal{M}(P)$. Moreover, when $\mathrm{Supp}(P)$ is finite, Whitmeyer and Whitmeyer [2021] show that any extreme point Q satisfies $|\mathrm{Supp}(Q)| \leq |\mathrm{Supp}(P)|$.[12] Thus, without loss, an optimal distribution $Q^\star$ can be chosen to have finite support.

3.2. **Informed sender**

In this section, we examine the dynamic game with an informed sender—one who knows the data-generating process. We show that when the process is stationary and ergodic, calibrated strategies induce mean-preserving contractions of the distribution of conditionals. This reduces the sender's dynamic problem to a static persuasion problem.

Given a stochastic process μ and a forecasting strategy σ, define the *distribution of conditionals* and the *distribution of forecasts*, whenever the respective limits exist, by

$$C_\mu := \lim_{T\to\infty} \frac{1}{T}\sum_{t=1}^{T} \delta_{p_t}, \qquad F_{\mu,\sigma} := \lim_{T\to\infty} \frac{1}{T}\sum_{t=1}^{T} \delta_{f_t}. \tag{3.5}$$

where $p_t = \mu(\cdot \mid \omega_1, \ldots, \omega_{t-1})$, and f_t is drawn according to $\sigma(f_1, \omega_1, \ldots, f_{t-1}, \omega_{t-1})$. Let $\mathcal{F}_{\mu,\sigma}$ denote the set of all weak limit points of the empirical distributions of forecasts. Under truthful forecasting, the empirical distributions of forecasts and conditionals coincide. Both C_μ and $F_{\mu,\sigma}$ may depend on the realized history and are therefore random.

For any stochastic process μ, the truthful forecasting strategy is always calibrated. However, without further restrictions on the process, characterizing the set of feasible long-run outcomes that pass the calibration test is difficult since C_μ may fail to exist. We therefore restrict attention to stationary ergodic processes, for which C_μ exists and is constant almost surely (see Lemma 2 in Appendix A). For illustration, consider Example 1. The Markov chain is irreducible and aperiodic, with invariant distribution $(0.5, 0.5)$.

[11] $\mathrm{Co}(A)$ denotes the convex hull of A.

[12] For non-atomic priors on an interval, Dworczak and Martini [2019], Kleiner et al. [2021], and Arieli et al. [2023] show that the extreme points correspond to bi-pooling signaling policies.

Thus, rain and no rain each occur with long-run frequency 0.5, and the corresponding conditionals are 0.8 and 0.2, respectively. Hence, C_μ places equal mass on 0.8 and 0.2. This distribution serves as the prior in the associated persuasion problem.

We now state our main result. The maximum payoff of the informed sender in the dynamic forecasting game equals the persuasion value with prior C_μ.

Theorem 1. *For a stationary ergodic process μ, the informed sender's maximum payoff equals* $\mathrm{Per}(C_\mu)$.

The key step in proving Theorem 1 is to characterize the distributions of forecasts that can arise under calibration.

Lemma 1. *Fix a stationary ergodic process μ. If a forecasting strategy σ is calibrated, then $\mathcal{F}_{\mu,\sigma} \subseteq \mathcal{M}(C_\mu)$. Conversely, under Assumption 1, for any $Q \in \mathcal{M}(C_\mu)$ with finite support, there exists a calibrated strategy σ such that $F_{\mu,\sigma} = Q$ and that fails the calibration test with expected long-run frequency zero.*

Intuitively, calibration requires the sender to use appropriate weights to combine conditionals into forecasts whose long-run frequencies of states match the announced forecasts. These weights correspond precisely to a mean-preserving contraction of the distribution of conditionals.

Proof sketch of Theorem 1. The proof has two parts. We first show that no forecasting strategy can give the sender more than the persuasion value, and then construct a strategy that attains it.

For the upper bound, suppose first that the sender always passes the calibration test. The receiver then best responds to each announced forecast f. Since the error margins converge to zero, the calibration error vanishes in the long run. Thus, the sender's average payoff approaches the frequency-weighted average of her indirect utility $\hat{u}_S(f)$. By Lemma 1, every limit point of the empirical distribution of forecasts is a mean-preserving contraction of C_μ. Therefore, the sender's payoff cannot exceed the persuasion value $\mathrm{Per}(C_\mu)$.

It remains to consider whether the sender can do better by occasionally failing the calibration test. Such failures may be beneficial because they can change the distribution of states in the periods in which the sender passes, making those periods more favorable to her. The key step is to show that any such gain is bounded proportionally to the frequency of failures. Since each failure incurs the punishment cost c, Assumption 2 ensures that these costs more than offset the potential gain. Hence, allowing the sender to fail the test cannot raise her long-run payoff above the persuasion value. Moreover, any optimal forecasting strategy must fail the calibration test in a vanishing fraction of periods in expectation.

To attain the bound, let $Q^\star \in \mathcal{M}(C_\mu)$ be a finitely supported distribution that attains the persuasion value, and let $\pi^\star$ be a signaling policy that implements it. Stationarity

and ergodicity imply that the conditionals occur with long-run distribution C_μ. In every period, after observing the current conditional p_t, the sender draws her forecast according to $\pi^\star(\cdot \mid p_t)$. Thus, in the long run, the dynamic forecasting strategy reproduces the distribution of conditionals and forecasts generated by $\pi^\star$ in the static persuasion problem, and hence induces $Q^\star$. By Lemma 1 and Assumption 1, this strategy is calibrated and fails the calibration test in a vanishing fraction of periods in expectation. Its expected long-run payoff is therefore $\mathrm{Per}(C_\mu)$. □

More generally, the restriction imposed by calibration extends beyond stationary ergodic processes. For an arbitrary stochastic process, the empirical distributions of forecasts and conditionals need not converge, so their long-run distributions may fail to exist. Nevertheless, when the sender uses finitely many forecasts, calibration still imposes an analogous restriction directly on their empirical joint frequencies; see Proposition 5 in Appendix A.

3.3. **Uninformed sender**

In this section, we examine the dynamic game with an uninformed sender. A seminal result of Foster and Vohra [1998] shows that even an uninformed sender can construct a calibrated forecasting strategy for any stochastic process. We derive the payoff she can guarantee and compare it with that of an informed sender. We first consider an adversarial environment in which nature chooses the sequence of states to minimize the sender's payoff, and then analyze the case where the process is a fixed Markov chain that remains unknown to the sender.

For this section, we focus on a less ambitious goal: ϵ-calibration.[13] This relaxation allows us to restrict attention to a finite set of forecasts. Moreover, rather than applying the finite calibration test in every period, we impose ϵ-calibration as an asymptotic requirement and let the receiver play $a_t = \hat{a}(f_t)$ in every period.[14]

Given $\epsilon > 0$, let $F_\epsilon \subseteq \Delta(\Omega)$ be a finite ϵ-grid.[15] For each $p \in \Delta(\Omega)$, let

$$f^*(p) \in \arg\min_{f \in F_\epsilon} \|p - f\|.$$

Thus, $f^*(p)$ is a forecast in the grid closest to p, and satisfies $\|p - f^*(p)\| \leq \epsilon$. If there are several such forecasts, choose one that minimizes the sender's expected payoff at p.

[13] This weaker criterion is common in the literature. In fact, using ϵ-calibrated strategies and the "doubling trick" one can obtain calibrated strategies [Cesa-Bianchi and Lugosi, 2006, Mannor and Stoltz, 2010, Perchet, 2014].

[14] In the informed sender case, the error margins are tailored to a fixed stationary ergodic process. Here the process is unknown and, in the adversarial case, nature chooses states strategically, so we focus on asymptotic ϵ-calibration.

[15] Such finite discretizations of $\Delta(\Omega)$ are standard in the calibration literature (see, e.g., Foster and Vohra, 1998, Perchet, 2014, Foster and Hart, 2021, 2023, Hart, 2025).

For each $p \in \Delta(\Omega)$, define the sender's approximate indirect utility

$$\hat{u}_S^\epsilon(p) := \sum_{\omega \in \Omega} p(\omega) u_S(\omega, \hat{a}(f^*(p))).$$

Thus, $\hat{u}_S^\epsilon(p)$ is the sender's expected payoff when the state distribution is p and the receiver best responds to the forecast $f^*(p)$. The grid approximation therefore implies that $|\hat{u}_S^\epsilon(p) - \hat{u}_S(f^*(p))| \leq L\epsilon$.

We begin with the adversarial case in which nature chooses the sequence of states to minimize the sender's payoff. In this environment, nature may choose the state, possibly at random, in each period based on the entire history of past forecasts and states.

Since we do not assume that states are drawn according to any fixed stochastic process, we adopt a pathwise performance criterion and evaluate the sender's payoff against a benchmark determined by the realized empirical distribution of states. Let $\overline{\omega}_T \in \Delta(\Omega)$ denote the *empirical distribution* of states up to period T:

$$\overline{\omega}_T := \frac{1}{T} \sum_{i=1}^{T} \delta_{\omega_i}.$$

The following notion of attainability formalizes this pathwise benchmark and is adapted from the online learning literature (see, e.g., Mannor et al., 2009, Bernstein et al., 2014).

Definition 5. A function $h \colon \Delta(\Omega) \to \mathbb{R}$ is *attainable* for the uninformed sender if there exists a forecasting strategy σ such that, for any nature's strategy τ, the following conditions hold $\mathbb{P}_{\sigma,\tau}$-almost surely:

$$(1)\ \liminf_{T \to \infty} \left(\frac{1}{T} \sum_{t=1}^{T} u_S(\omega_t, a_t) - h(\overline{\omega}_T) \right) \geq 0,$$

$$(2)\ \limsup_{T \to \infty} \sum_{f \in F_\epsilon} \frac{N_T[f]}{T} \left\| \overline{\omega}_T[f] - f \right\| \leq \epsilon.$$

The first condition requires the sender's average payoff to asymptotically exceed $h(\overline{\omega}_T)$. The second condition requires that the forecasting strategy be ϵ-calibrated.

We show that the closed convex hull of the sender's approximate indirect utility is attainable. Let $\overline{\mathrm{Co}}\, g$ denote the closed convex hull of a function g.[16]

Theorem 2. *The function $\overline{\mathrm{Co}}\, \hat{u}_S^\epsilon$ is attainable for the uninformed sender.*

This bound is the finite-grid analogue of the sender's worst payoff in the canonical Bayesian persuasion problem as a function of the prior.

Proof sketch of Theorem 2. The result follows from Blackwell approachability. We define a vector-valued payoff that tracks the sender's payoff, the calibration error, and the empirical distribution of states. We then define a closed and convex target set requiring

[16] Given a function $g : X \to \mathbb{R}$, over a convex domain X, its closed convex hull is the function whose epigraph is $\overline{\mathrm{Co}}(\{(x, r) : r \geq g(x)\})$ where $\overline{\mathrm{Co}}(X)$ is the closed convex hull of the set X.

the sender to obtain at least $\overline{\mathrm{Co}}\,\hat{u}_S^\epsilon(p)$, where p denotes the empirical distribution of states, while remaining ϵ-calibrated.

Blackwell's dual condition requires that, for every $p \in \Delta(\Omega)$, there exists a (randomized) forecast $x \in \Delta(F_\epsilon)$ whose expected vector payoff belongs to the target set. To verify this condition, fix any p and choose the pure forecast $f^*(p)$. The calibration error is at most ϵ by construction of the grid, and the sender's expected payoff is $\hat{u}_S^\epsilon(p) \geq \overline{\mathrm{Co}}\,\hat{u}_S^\epsilon(p)$. Hence, Blackwell's condition holds and the target set is approachable. Moreover, the corresponding approachability strategy provides an explicit forecasting strategy for the uninformed sender. □

We now consider a non-adversarial environment, where states are governed by an unknown ergodic Markov chain of order k (i.e., the next state depends only on the last k states). In this case, the uninformed sender can guarantee a higher payoff while remaining ϵ-calibrated.

Proposition 1. *Suppose μ is an irreducible and aperiodic Markov chain of order k. Then the uninformed sender can guarantee a payoff of at least $\hat{u}_S^\epsilon(m(C_\mu))$ using an ϵ-calibrated forecasting strategy.*

In the adversarial case, nature can vary the empirical distribution of states across blocks while preserving the same overall distribution. This can reduce the payoff the sender can guarantee while remaining calibrated. In contrast, for an ergodic Markov chain, the empirical distributions over long blocks approach the same invariant distribution, ruling out such strategic variation. This allows the sender to guarantee $\hat{u}_S^\epsilon(m(C_\mu))$, the finite-grid analogue of the no disclosure payoff in the persuasion problem with prior C_μ. The sender need not know in advance that the process is a Markov chain. If nature's play is favorable, she attains the higher benchmark; otherwise she secures the adversarial guarantee.[17]

In summary, an uninformed sender can attain the function $\overline{\mathrm{Co}}\,\hat{u}_S^\epsilon$ in the adversarial case and guarantee $\hat{u}_S^\epsilon(m(C_\mu))$ when the process is an ergodic Markov chain. Together, Theorem 1 and Proposition 1 provide a benchmark for the value of information in Markov environments.

3.4. **Application: Financial Platform**

We consider a financial forecasting platform that provides probability forecasts of a binary market state to a user. The platform's credibility is evaluated through calibration, and its payoff from a forecast depends on its precision and user engagement.

A financial platform (sender) provides daily probability forecasts about a binary market state $\omega_t \in \{H, L\}$. After observing the forecast, a user (receiver) decides whether to invest in the market. The user prefers to invest if and only if the state is H.

[17] Bernstein et al. [2014] establish this property in more general approachability settings.

The state evolves according to a Markov chain with transition probabilities $\mu(\omega_{t+1} = H \mid \omega_t = H) = \mu(\omega_{t+1} = L \mid \omega_t = L) = 0.95$. The initial state is drawn from the invariant distribution, which assigns equal probability to L and H. Thus, the process is stationary and ergodic and induces the distribution of conditionals C_μ that places equal mass on $\{0.05, 0.95\}$.

To illustrate the calibration restriction, we allow the platform's payoff to depend directly on its forecast. This reduced-form utility has two components. First, the platform benefits from user engagement, interpreted as time spent on the platform. Engagement is highest for coarse forecasts and declines as forecasts become extreme. The platform's utility from user engagement following forecast f (see Fig. 1) is given by:

$$\hat{u}_S^{\text{eng}}(f) = \begin{cases} C(G_f) & \text{if } 0.05 \leq f \leq 0.95, \\ 0 & \text{otherwise,} \end{cases} \tag{3.6}$$

where G_f is a distribution of beliefs with support $\{0.05, 0.95\}$ and mean f, $C(G_f) = \kappa_1\left(\int_0^1 \ell(q)\, dG_f(q) - \ell(f)\right)$ with $\kappa_1 > 0$, and $\ell(q) = q\log(\frac{q}{1-q}) + (1-q)\log(\frac{1-q}{q})$.[18] Here, $C(G_f)$ measures the additional information the user acquires after observing forecast f: coarser forecasts induce more information acquisition and hence greater engagement.

Second, the platform derives utility from announcing precise forecasts, which we interpret as gain in reputation. The reputational utility from announcing forecast f (see Fig. 2) is given by

$$\hat{u}_S^{\text{rep}}(f) = \kappa_2(f - 0.5)^2, \tag{3.7}$$

for some constant $\kappa_2 > 0$.

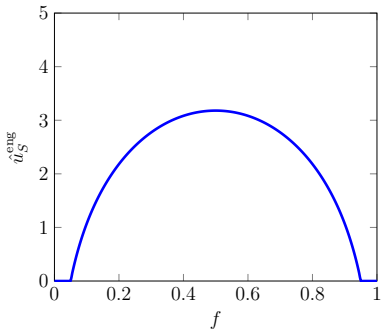


FIGURE 1. $\hat{u}_S^{\text{eng}}(f)$

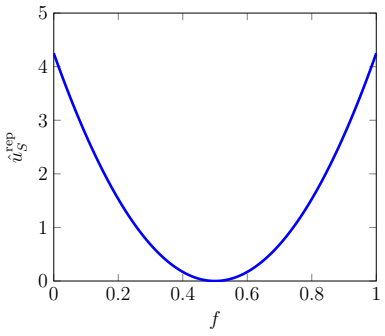


FIGURE 2. $\hat{u}_S^{\text{rep}}(f)$

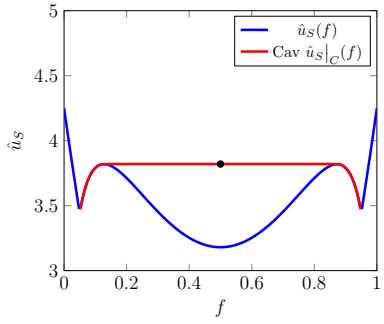


FIGURE 3. $\hat{u}_S(f)$

The platform's total indirect utility from announcing forecast f (see Fig. 3) is therefore

$$\hat{u}_S(f) = \hat{u}_S^{\text{eng}}(f) + \hat{u}_S^{\text{rep}}(f). \tag{3.8}$$

[18]The user engagement utility can be microfounded via an information acquisition problem as in Morris and Strack [2019]. Starting from a prior belief f, the user acquires information until the belief crosses a 95% confidence threshold. Then G_f denotes the resulting distribution of posteriors, and the associated cost $C(G_f)$ is proportional to the expected change in the log-likelihood ratio.

The platform faces a trade-off: revealing more precise forecasts increases reputational utility but reduces user engagement. The platform's objective is to choose a calibrated forecasting strategy that maximizes its expected long-run average payoff.

Since $\mathrm{Supp}(C_\mu) = \{0.05, 0.95\}$ is affinely independent, Lemma 1 and (3.4) imply that the platform's maximum payoff among calibrated strategies is given by

$$\mathrm{Per}(C_\mu) = \mathrm{Cav}\, \hat{u}_S\big|_{[0.05,0.95]}(0.5). \tag{3.9}$$

The black dot at $f = 0.5$ in Figure 3 illustrates this value, where $C = [0.05, 0.95]$. For suitable values of κ_1 and κ_2, the distribution of conditionals C_μ and an optimal distribution of forecasts $Q^\star$ are given by:

$$C_\mu = \begin{bmatrix} \lambda \\ p \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{5}{100} & \frac{95}{100} \end{bmatrix} \qquad Q^\star = \begin{bmatrix} \nu \\ q \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{15}{100} & \frac{85}{100} \end{bmatrix}$$

The distribution $Q^\star$ can be induced by an optimal forecasting strategy $\sigma^\star$, where for any $t \in \mathbb{N}$:

$$\sigma^\star(f_t = 15\% \mid p_t = 5\%) = \frac{8}{9}, \qquad \sigma^\star(f_t = 85\% \mid p_t = 5\%) = \frac{1}{9},$$
$$\sigma^\star(f_t = 15\% \mid p_t = 95\%) = \frac{1}{9}, \qquad \sigma^\star(f_t = 85\% \mid p_t = 95\%) = \frac{8}{9}.$$

The optimal calibrated strategy announces the coarse forecasts $f_t \in \{15\%, 85\%\}$ rather than the truthful conditionals $p_t \in \{5\%, 95\%\}$. This corresponds to the concave envelope of $\hat{u}_S$ restricted to the $\mathrm{Co}(\mathrm{Supp}(C_\mu))$ (red line in Fig. 4). More generally, as the platform places greater weight on reputation relative to engagement, its optimal forecasts become more precise. Calibration limits how far this can go: although the platform would prefer to issue extreme forecasts of 0% and 100%, these cannot occur with positive long-run frequency, since the forecast distributions must be supported on $\mathrm{Co}(\mathrm{Supp}(C_\mu))$.

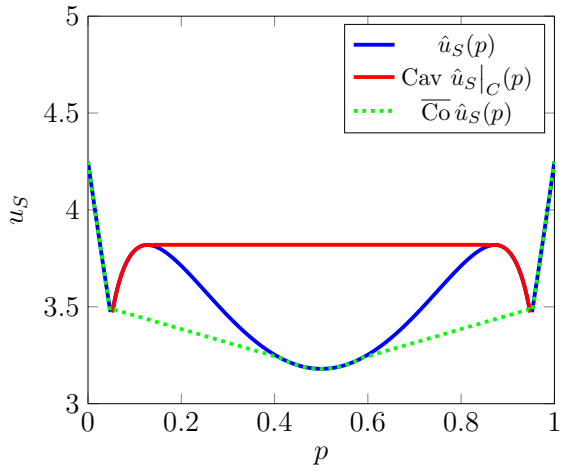


FIGURE 4. Financial platform guarantees: (i) Informed sender (stationary, ergodic) - red line, (ii) Uninformed sender (adversarial) - green dotted line, (iii) Uninformed sender (ergodic Markov chain) - blue line.

For any stochastic process, truthful forecasting is calibrated. Whenever C_μ is well defined, truthful forecasting achieves $\mathbb{E}_{C_\mu}[\hat{u}_S]$. In the associated persuasion problem with prior C_μ, this corresponds to the full disclosure.

In the adversarial case, an uninformed platform can approximately guarantee the closed convex hull of the indirect utility, $\overline{\mathrm{Co}}\,\hat{u}_S$, evaluated at the realized empirical distribution of states (green dotted line). If the process is an ergodic Markov chain, this guarantee improves to $\hat{u}_S(m(C_\mu))$ (blue line), again up to the grid approximation. The gap between the red and blue lines at $m(C_\mu)$ compares the informed sender's optimal payoff with the payoff guarantee for the uninformed sender, providing a benchmark for the value of information about the Markov chain.

4. Forecasting against regret minimizers

In this section, we consider a receiver who uses regret minimization, instead of the calibration test, as the heuristic for decision-making. Regret is an exogenous criterion to evaluate a strategy in non-Bayesian environments [Cesa-Bianchi and Lugosi, 2006]. It measures the difference between the receiver's realized average payoff and what he could have obtained by repeatedly playing a fixed action. Regret-minimizing strategies ensure that, in the long run, no fixed action would have resulted in a higher average payoff. Calibration and regret minimization are closely related notions in the online learning literature [Perchet, 2014]. We use this connection to compare strategic forecasting under the two decision rules.

Apart from this change in the receiver's behavior, the environment and timing are as in Section 2: in each period the sender sends a forecast, the receiver chooses an action, the state is realized, and the game moves to the next period.

We establish two main results. First, we show that a receiver who myopically best responds to calibrated forecasts incurs no regret in the long run. Second, focusing on mean-based learning algorithms, we show that the sender can attain the calibration benchmark and in some cases strictly more.

We evaluate regret conditional on each forecast, i.e., contextual regret with the forecast as the context. Intuitively, the receiver has no regret with respect to a forecast f if, on the set of periods in which f is sent, he cannot improve his long-run average payoff by switching to any fixed action $a^* \in A$. Let $\overline{u}_{R,T}[f]$ denote the receiver's average payoff up to period T conditional on f:[19]

$$\overline{u}_{R,T}[f] := \frac{1}{N_T[f]} \sum_{t=1}^{T} \mathbf{1}_{\{f_t = f\}} u_R(\omega_t, a_t).$$

Definition 6. The receiver has no regret with respect to forecast f if

$$\limsup_{T\to\infty} \frac{N_T[f]}{T} \Big(\max_{a^* \in A} u_R(\overline{\omega}_T[f], a^*) - \overline{u}_{R,T}[f] \Big) \leq 0 \quad \text{almost surely.} \tag{4.1}$$

[19] If $N_T[f] = 0$, set $\overline{u}_{R,T}[f] = 0$.

The next proposition shows that if the receiver best responds to the forecasts of a calibrated strategy, that is, plays $a_t = \hat{a}(f_t)$ in every period, then he incurs no regret with respect to any forecast. This provides a justification for using calibration as a heuristic for decision-making.

Proposition 2. *Given a calibrated forecasting strategy, a receiver who myopically best responds to every forecast has no regret with respect to any forecast.*

For a stationary ergodic process, Theorem 1 characterizes the maximum payoff the informed sender can achieve under the calibration test. We now consider a receiver who instead uses a learning algorithm.

We follow Braverman et al. [2018] and Deng et al. [2019] and focus on the natural class of *mean-based learning algorithms*. Mean-based algorithms assign vanishing probability to actions whose cumulative payoff is sufficiently below that of another action. Intuitively, mean-based strategies play the action that historically performs the best. This class includes standard no-regret algorithms such as Multiplicative Weights, Follow-the-Perturbed-Leader, and EXP3.

We consider the T-period version of the game, where T is large.[20] We analyze the sender's expected average payoff over the T periods as $T \to \infty$.

Definition 7. Let $U^f_{a,t} = \sum_{i=1}^{t} \mathbf{1}_{\{f_i = f\}} u_R(\omega_i, a)$ be the receiver's cumulative payoff from action a over the first t periods when the forecast was f. An algorithm is γ-mean-based if, for every forecast f, whenever $U^f_{a,t} < U^f_{b,t} - \gamma T$ for some $b \in A$, the probability of choosing action a following forecast f in period $t+1$ is at most γ. An algorithm is mean-based if it is γ-mean-based for some $\gamma = o(1)$.

The following proposition shows that the sender can attain the same benchmark as under the calibration test. To rule out degenerate ties, we assume that, within $\mathrm{Co}(\mathrm{Supp}(C_\mu))$, every receiver best response can be made strict by an arbitrarily small perturbation of the belief.[21]

Proposition 3. *For a stationary ergodic process μ and a mean-based receiver, the sender can guarantee* $\mathrm{Per}(C_\mu)$.

Recall that $\pi^\star$ is an optimal signaling policy in the static persuasion problem which uses only finitely many forecasts. As in Theorem 1, the sender can use $\pi^\star$ in each period to generate its optimal distribution of forecasts. If the receiver has a unique best response to each forecast, that action accumulates a linear payoff advantage over every other action, since the empirical distribution of states following the forecast converges to it. A mean-based receiver therefore chooses these best responses in all but a vanishing fraction of periods, giving the sender the persuasion value. When best responses are tied, a small

[20] An infinite horizon analogue can be obtained by using, in each period t, the threshold $\gamma_t t$, where $\gamma_t \to 0$. We focus on the finite horizon formulation used by Braverman et al. [2018] for simplicity.

[21] For finite state and action spaces, this condition holds generically in the receiver's payoff function.

perturbation of the optimal signaling policy makes the sender-preferred actions strictly optimal at an arbitrarily small payoff cost.

We have shown that the sender can always do at least as well against a mean-based receiver as under the calibration benchmark. The next theorem shows that the sender can sometimes do strictly better.

Theorem 3. *There exists a stationary ergodic process μ and a game such that, against a mean-based receiver, the sender can guarantee a payoff strictly greater than* $\mathrm{Per}(C_\mu)$.

Proof. Consider the following payoff matrix, which represents $u_S(\omega, a)$ and $u_R(\omega, a)$:

	a_1	a_2	a_3	a_4
0	$(2, 8)$	$(0, 7)$	$(4, 3)$	$(2, 0)$
1	$(2, 0)$	$(4, 3)$	$(0, 7)$	$(2, 8)$

The receiver's best response depends on his belief $p = \mathbb{P}(\omega = 1)$. It is optimal to play a_1 for $p \in [0, 0.25]$, a_2 for $p \in [0.25, 0.5]$, a_3 for $p \in [0.5, 0.75]$, and a_4 otherwise (see Fig. 5).

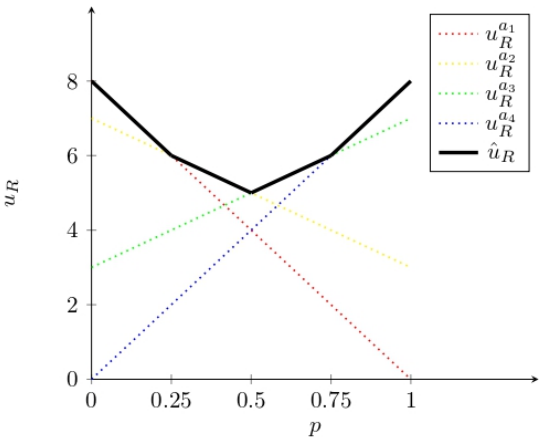


FIGURE 5. Receiver's indirect utility: $\hat{u}_R$

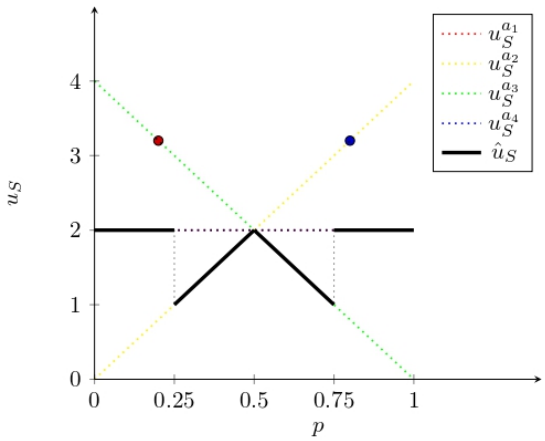


FIGURE 6. Sender's indirect utility: $\hat{u}_S$

The state evolves according to a Markov chain, initialized at its invariant distribution, with transition probabilities $\mu(\omega_{t+1} = 1 \mid \omega_t = 1) = 0.8$ and $\mu(\omega_{t+1} = 1 \mid \omega_t = 0) = 0.2$. Hence, the distribution of conditionals C_μ has equal mass on 20% and 80%. An optimal signaling policy in the persuasion problem is full or no disclosure, which implies $\mathrm{Per}(C_\mu) = 2$ (see Fig. 6).

We now describe a strategy that allows the sender to guarantee a strictly higher payoff against a mean-based receiver. The construction exploits the fact that mean-based algorithms adapt gradually over time.

Because the receiver uses a mean-based learning algorithm, his behavior is driven by cumulative payoffs. Conditional on forecast f, the cumulative payoff from action a up to period t is $U_{a,t}^f = N_t[f] \cdot u_R(\overline{\omega}_t[f], a)$. Thus, actions that yield higher payoff under the empirical distribution $\overline{\omega}_t[f]$ accumulate larger cumulative payoffs. Hence, once an action's

cumulative payoff exceeds that of every other action by more than γT, the receiver plays it with high probability.

The game has two phases, and the sender uses only two forecasts: l and h. During the first $\frac{T}{2}$ periods, the sender announces l when $p_t = 20\%$ and h when $p_t = 80\%$. Under a mean-based learning algorithm, the receiver plays a_1 after l and a_4 after h with high probability. For large T, the sender's average payoff in this first phase is approximately 2.[22]

In the remaining $\frac{T}{2}$ periods, the sender swaps the forecasts: she announces l when $p_t = 80\%$ and h when $p_t = 20\%$. After this change, the empirical frequency given l gradually increases from 20% toward 50%. Once it exceeds 25%, action a_2 becomes more profitable than a_1, and the receiver switches to a_2 with high probability. A similar argument applies to h: as the empirical frequency decreases, the receiver eventually switches from a_4 to a_3.

For most of this second phase, the receiver plays a_2 and a_3 when the conditional distributions are 80% and 20%, respectively. These actions are not best responses to the conditional beliefs and therefore result in payoffs outside the graph of indirect utilities (see the red and blue dots in Fig. 6). A direct calculation shows that the sender's average payoff in this phase is approximately $\frac{34}{11}$. Combining both phases, the overall average payoff is approximately $\frac{28}{11} > 2$. Thus, the sender achieves strictly more than the calibration benchmark. □

A natural extension is to the case in which the receiver does not observe the state $(\omega_t)_{t\geq 1}$ and observes only the payoff from the chosen action. Standard techniques from bandit learning, such as inverse propensity score estimators, can be used to estimate the payoffs of unplayed actions with vanishing exploration costs [Bubeck and Cesa-Bianchi, 2012, Lattimore and Szepesvári, 2020].

5. Related Literature

Strategic Information Transmission: Our work contributes to the literature on communication between an informed sender and an uninformed receiver. In cheap talk [Crawford and Sobel, 1982], the sender's message is unverifiable, whereas in Bayesian persuasion [Kamenica and Gentzkow, 2011], the sender commits to how the message is generated. In our dynamic setup, the sender does not have commitment power and the messages are probabilistic forecasts disciplined by calibration. In particular, we contribute to the study of communication in dynamic environments (Renault et al., 2013, Margaria and Smolin, 2018, Kuvalekar et al., 2022). More specifically, the question of whether repetition can substitute for commitment is explored in Pei [2023], Best and Quigley [2024], Mathevet et al. [2026], and Jain [2026], who offer rationales based on lying

[22] Since $\gamma = o(1)$ and the cumulative payoff difference grows at a positive rate, the additional time needed to exceed γT is negligible and does not affect the limiting average payoff.

costs, coarse summaries, reputation and quota mechanisms, respectively. In contrast, we consider a general dynamic setting in which the receiver evaluates the sender using a calibration test. Our main result shows that the dynamic forecasting game can be reduced to a static persuasion problem with a simplex state space in which the sender's utility depends only on the posterior mean (Gentzkow and Kamenica, 2016, Kolotilin, 2018, Dworczak and Martini, 2019, Kleiner et al., 2021, Arieli et al., 2023). Thus, our work shows how calibration can generate the Bayesian persuasion benchmark. Our paper is also related to the static forecasting model of Guo and Shmaya [2021]. As in our model, messages are probabilistic forecasts. However, they define calibration error as the distance between a forecast and the true conditional distribution over states induced by the sender's strategy, and impose an exogenous cost based on this deviation. Unlike their equilibrium framework, our receiver evaluates forecasts through a calibration test based on realized forecasts and states, without forming beliefs about the sender's strategy. This restricts feasible long-run forecast distributions and leads to the persuasion benchmark.

Calibration and Expert Testing: The initial focus of this literature has been to design a statistical test that can distinguish between an informed expert, who knows the data-generating process, and an uninformed expert, who does not. The calibration test is an objective and popular criterion for evaluating experts. However, Foster and Vohra [1998] show that even an uninformed expert can produce calibrated forecasts for any process. Despite this, calibration is crucial for decision-making based on forecasts [Foster and Hart, 2021]. We build on this literature by exploring the use of the calibration test as a heuristic for decision-making. Specifically, we characterize the extent of strategic forecasting when the calibration test is used to determine the credibility of the expert. We compare the payoffs that an informed and an uninformed expert can achieve for a given process. Related work studies tests of expert forecasts in decision problems, including Echenique and Shmaya [2007], Gradwohl and Salant [2011] and Olszewski and Pęski [2011]. These papers examine how decision-makers can benefit from expert advice while being protected against uninformed forecasts. Calibration also has important applications in machine learning (see Gupta and Ramdas, 2021). We refer curious readers to Foster and Vohra [2013] and Olszewski [2015] for comprehensive surveys on this topic.

Regret minimization and Online Learning: Regret minimization, which uses the best fixed action in hindsight as a natural benchmark, provides simple procedures for online decision-making and ensures behavior that leads to as-if rational outcomes (see Cesa-Bianchi and Lugosi, 2006, Hart and Mas-Colell, 2013). There has been growing interest in studying regret minimization within strategic environments. We focus on optimizing against regret minimizing agents [Braverman et al., 2018, Deng et al., 2019, Camara et al., 2020, Lin and Chen, forthcoming]. The closest paper is Lin and Chen

[forthcoming], which independently studies a repeated sender-receiver setting and provides non-asymptotic bounds on what the sender can guarantee. They analyze an i.i.d. setting where the sender observes the realized state in each period. In contrast, we allow dependence across periods, and the sender observes only the conditional distribution of the current state. Consequently, their bounds and techniques do not directly apply to our setup. The class of *mean-based learning algorithms* was introduced by Braverman et al. [2018] and further explored by Deng et al. [2019]. A common theme in the literature is that the sender can obtain higher payoff than the rational benchmark when facing such algorithms. Similarly, we show the sender can obtain a higher utility than the calibration benchmark against a mean-based learner. Additionally, *U-calibration*, as introduced in Kleinberg et al. [2023], is relevant as it ensures low external regret for receivers with unknown payoff functions.

6. Conclusion

We study strategic forecasting when credibility is determined by calibration. Calibration constrains, but does not eliminate, the sender's ability to forecast strategically. For stationary and ergodic processes, calibration allows the sender to strategically coarsen truthful forecasts: the distributions that can arise under calibration are mean-preserving contractions of the distribution of truthful forecasts. This reduces the sender's dynamic optimization problem to a static persuasion problem. Comparing informed and uninformed senders further separates credibility from information: an expert need not know the data-generating process to produce calibrated forecasts. However, information allows the sender to benefit more from strategic forecasting. Finally, our results on regret minimization show that replacing calibration with a different decision-making heuristic does not necessarily eliminate the sender's ability to benefit from strategic forecasts.

Many problems remain open for the setting that we study. In particular, characterizing the optimal calibrated strategy for an arbitrary stochastic process remains open. Since this problem may be intractable in full generality, a natural question is for which classes of stochastic processes one can find a calibrated strategy that does better than truthful forecasting. A complete characterization of attainable payoffs against no-regret learners also remains open.

The model admits natural extensions. The receiver may use alternative statistical tests to assess credibility. This raises the question of which tests make truthful forecasting optimal and which limit the scope for strategic misreporting. One can also let the receiver's actions affect state transitions—for example, in a Markov decision process. An interesting question then is to characterize which distributions of forecasts are feasible under calibration in that setting.

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Appendix A. Omitted Results and Proofs

A.1. Proposition 4

Proposition 4. *Fix a stationary ergodic process μ. There exists a sequence $(\epsilon_t)_{t\geq 1}$ with $\epsilon_t \to 0$ such that every calibrated strategy induced by a fixed rule $\pi : D \to \Delta(F_0)$, with F_0 finite, fails the calibration test only finitely many times almost surely.*

Proof. Fix a calibrated forecasting strategy σ of the stated form, and let F_0 denote its finite forecast support. For each $f \in F_0$,

$$\frac{N_T[f]}{T}\big(\overline{\omega}_T[f] - f\big) = \frac{1}{T}\sum_{t=1}^{T} \mathbf{1}_{\{f_t=f\}}(\delta_{\omega_t} - p_t) \tag{A.1}$$

$$+ \frac{1}{T}\sum_{t=1}^{T}\big(\mathbf{1}_{\{f_t=f\}} - \pi(f \mid p_t)\big)(p_t - f) \tag{A.2}$$

$$+ \frac{1}{T}\sum_{t=1}^{T}\pi(f \mid p_t)(p_t - f). \tag{A.3}$$

The first two terms are averages of bounded martingale differences and converge to zero almost surely. Since σ is calibrated, the norm of the left-hand side also converges to zero for every $f \in F_0$. Hence, the third term converges to zero almost surely. Since μ is stationary and ergodic, Lemma 2 implies that $N_T[p]/T \to C_\mu(p)$ almost surely for every $p \in D$. It follows that

$$\sum_{p\in D} C_\mu(p)\pi(f \mid p)(p - f) = 0 \qquad \text{for every } f \in F_0. \tag{A.4}$$

By Lemma 3 of Foster et al. [2011], applied to the two bounded martingale-difference sequences above, there exist constants $z_1, z_2 > 0$ such that, for every $T \geq 1$, every $f \in F_0$,

and every $\theta > 0$,

$$\mathbb{P}\left(\left\|\frac{1}{T}\sum_{t=1}^{T}\mathbf{1}_{\{f_t=f\}}(\delta_{\omega_t}-p_t)\right\|\geq\theta\right)\leq 2e^{-z_1T\theta^2},$$

and

$$\mathbb{P}\left(\left\|\frac{1}{T}\sum_{t=1}^{T}\big(\mathbf{1}_{\{f_t=f\}}-\pi(f\mid p_t)\big)(p_t-f)\right\|\geq\theta\right)\leq 2e^{-z_2T\theta^2}.$$

Let $z := \min\{z_1, z_2\}$, and let M_T denote the sum over $f \in F_0$ of the norms of the first two terms in (A.1)–(A.2). If $M_T > T^{-1/4}$, then at least one of these $2|F_0|$ nonnegative terms must exceed $T^{-1/4}/(2|F_0|)$. Hence, by a union bound,

$$\mathbb{P}\left(M_T > T^{-1/4}\right) \leq 4|F_0|\exp\left(-\frac{z\sqrt{T}}{4|F_0|^2}\right).$$

The right-hand side is summable in T. Therefore, by the Borel–Cantelli lemma,

$$M_T \leq T^{-1/4} \tag{A.5}$$

for all sufficiently large T, almost surely.

Next, define

$$R_T := \sum_{p\in D}\left|\frac{N_T[p]}{T}-C_\mu(p)\right|.$$

Since μ is stationary and ergodic, Lemma 2 implies that $R_T \to 0$ almost surely. Hence, there exists a deterministic sequence $b_T \to 0$, depending only on μ, such that

$$R_T \leq b_T \tag{A.6}$$

for all sufficiently large T, almost surely.[23]

Using (A.4), for every $f \in F_0$,

$$\begin{aligned}\frac{1}{T}\sum_{t=1}^{T}\pi(f\mid p_t)(p_t-f) &= \sum_{p\in D}\frac{N_T[p]}{T}\pi(f\mid p)(p-f)\\ &= \sum_{p\in D}\left(\frac{N_T[p]}{T}-C_\mu(p)\right)\pi(f\mid p)(p-f).\end{aligned}$$

Since $\|p-f\| \leq \sqrt{2}$ and $\sum_{f\in F_0}\pi(f\mid p) = 1$, summing over $f \in F_0$ gives

$$\sum_{f\in F_0}\left\|\frac{1}{T}\sum_{t=1}^{T}\pi(f\mid p_t)(p_t-f)\right\| \leq \sqrt{2}\,R_T.$$

Therefore, by (A.1)–(A.3), the calibration error satisfies

$$\sum_{f\in F_0}\frac{N_T[f]}{T}\|\overline{\omega}_T[f]-f\| \leq M_T + \sqrt{2}\,R_T.$$

[23]To see this, for each k choose T_k such that $\mathbb{P}(\sup_{T\geq T_k} R_T > 1/k) \leq 2^{-k}$, with $T_k \to \infty$, and set $b_T = 1/k$ for $T_k \leq T < T_{k+1}$. The claim then follows from the Borel–Cantelli lemma.

Finally, set

$$\epsilon_T := T^{-1/4} + \sqrt{2}\, b_T.$$

Then $\epsilon_T \to 0$, and (A.5)–(A.6) imply that the calibration error is at most ϵ_T for all sufficiently large T, almost surely. Hence, σ fails the calibration test only finitely many times almost surely. □

A.2. **Proof of Lemma 1**

(i) First, we show that for a stationary ergodic process μ and any calibrated strategy σ, $\mathcal{F}_{\mu,\sigma} \subseteq \mathcal{M}(C_\mu)$. The idea is to group forecasts into finitely many bins. For each partition, calibration implies that the resulting finite distribution is a simple mean-preserving contraction of C_μ. Letting the bin size go to zero completes the proof.

For each period T, define the empirical frequencies

$$N_T[p] = \sum_{t=1}^{T} \mathbf{1}_{\{p_t = p\}}, \qquad N_T[f] = \sum_{t=1}^{T} \mathbf{1}_{\{f_t = f\}}, \qquad N_T[f,p] = \sum_{t=1}^{T} \mathbf{1}_{\{f_t = f,\, p_t = p\}}.$$

Fix in advance a sequence of finite partitions of the set of forecasts whose maximum cell diameters converge to zero. Since there are only countably many cells, the martingale convergence used below holds simultaneously for all of them almost surely.

Let $Q \in \mathcal{F}_{\mu,\sigma}$ be any limit point of the empirical distribution of forecasts, and let (T_k) be a subsequence along which the empirical distribution converges to Q. Fix a partition $B_1, \ldots, B_m$ from this sequence, with maximum diameter η.

For each cell, let

$$N_T[B_j] = \sum_{f \in B_j} N_T[f], \qquad N_T[B_j, p] = \sum_{f \in B_j} N_T[f,p],$$

and define

$$G_T[p, B_j] = \frac{N_T[B_j, p]}{N_T[p]} \quad \text{if } N_T[p] > 0,$$

and arbitrarily otherwise. Then, for every B_j,

$$\sum_{p \in D} \frac{N_T[p]}{T} G_T[p, B_j] = \frac{N_T[B_j]}{T}. \tag{A.7}$$

It remains to verify mean preservation. Whenever $N_T[B_j] > 0$, let

$$\overline{f}_T[B_j] = \frac{1}{N_T[B_j]} \sum_{t : f_t \in B_j} f_t, \qquad \overline{p}_T[B_j] = \frac{1}{N_T[B_j]} \sum_{t : f_t \in B_j} p_t,$$

and define $\overline{\omega}_T[B_j]$ analogously. By the triangle inequality,

$$\frac{N_T[B_j]}{T} \|\overline{f}_T[B_j] - \overline{p}_T[B_j]\| \le \frac{N_T[B_j]}{T} \|\overline{f}_T[B_j] - \overline{\omega}_T[B_j]\| + \frac{N_T[B_j]}{T} \|\overline{\omega}_T[B_j] - \overline{p}_T[B_j]\|. \tag{A.8}$$

The first term converges to zero almost surely by calibration, since

$$\frac{N_T[B_j]}{T}\|\overline{f}_T[B_j] - \overline{\omega}_T[B_j]\| \leq \sum_{f\in B_j} \frac{N_T[f]}{T}\|\overline{\omega}_T[f] - f\| \to 0.$$

The second term can be rewritten as an average of bounded martingale differences,

$$\left\|\frac{1}{T}\sum_{t=1}^{T} X_t^j\right\|, \qquad X_t^j = \mathbf{1}_{\{f_t\in B_j\}}(\delta_{\omega_t} - p_t).$$

Hence, by the strong law of large numbers for bounded martingale differences, it also converges to zero. Thus, the left-hand side of (A.8) converges to zero almost surely.

Finally, since μ is stationary and ergodic, $N_T[p]/T \to C_\mu(p)$ almost surely. Since there are finitely many $p \in D$ and cells B_j, along a further subsequence we may assume that

$$G_{T_k}[p, B_j] \to G[p, B_j]$$

for every p and j. Since each G_{T_k} is row stochastic, so is G.

By (A.7), along (T_k), the empirical frequency of each cell converges to

$$q_j := \sum_{p\in D} C_\mu(p)G[p, B_j].$$

For every cell with $q_j > 0$, the average conditional $\overline{p}_{T_k}[B_j]$ converges, since the empirical frequencies of each pair (p, B_j) converge. By (A.8), the average forecast $\overline{f}_{T_k}[B_j]$ converges to the same limit; denote it by f_j^η.

Define

$$Q^\eta := \sum_{j:q_j>0} q_j \delta_{f_j^\eta}.$$

Taking to the limit in (A.7) and (A.8), G satisfies the mass- and mean-preservation conditions (3.1)–(3.2) for C_μ and Q^η. Hence, Q^η is a simple mean-preserving contraction of C_μ.

Along the chosen subsequence, Q^η is the limit of the empirical forecast distributions obtained by replacing forecasts within each cell by their average. Since each cell has diameter at most η, this changes each forecast by at most η. Therefore, $Q^\eta \to Q$ weakly as $\eta \to 0$. Hence, $Q \in \mathcal{M}(C_\mu)$, and therefore $\mathcal{F}_{\mu,\sigma} \subseteq \mathcal{M}(C_\mu)$.

(ii) Next, we show that if μ is stationary and ergodic, then for any $Q \in \mathcal{M}(C_\mu)$ with finite support, there exists a calibrated strategy σ that induces Q, that is, $F_{\mu,\sigma} = Q$ and fails the calibration test with expected long-run frequency zero.

Recall that the informed sender knows the distribution of conditionals C_μ. Fix $Q \in \mathcal{M}(C_\mu)$ and let π denote the signaling policy that implements Q. The informed sender can compute the conditional $p_t = \mu(\cdot \mid \omega^t)$ at the start of each period t. Define the forecasting strategy $\tilde{\sigma}$ by $\tilde{\sigma}_t(f_t = f \mid p_t = p) = \pi(f \mid p)$ for all t. We show the strategy $\tilde{\sigma}$

is calibrated. For any $f \in \text{Supp}(Q)$,

$$\frac{N_T[f]}{T}\|\overline{\omega}_T[f] - f\| \leq \left\|\frac{1}{T}\sum_{t=1}^{T}\mathbf{1}_{\{f_t=f\}}(\delta_{\omega_t} - p_t)\right\| + \left\|\frac{1}{T}\sum_{t=1}^{T}(\mathbf{1}_{\{f_t=f\}} - \pi(f \mid p_t))(p_t - f)\right\| + \left\|\frac{1}{T}\sum_{t=1}^{T}\pi(f \mid p_t)(p_t - f)\right\|.$$

The first two terms are averages of bounded martingale differences and therefore converge to zero almost surely. By stationarity and ergodicity, the third term converges almost surely to $\sum_p C_\mu(p)\pi(f \mid p)(p-f)$, which equals zero since π implements $Q \in \mathcal{M}(C_\mu)$.

Summing over the finite $f \in \text{Supp}(Q)$, we conclude that $\tilde{\sigma}$ is calibrated. Moreover, by Assumption 1, it fails the test only in expected long-run frequency zero of periods.

Finally, we show that $\tilde{\sigma}$ induces the distribution of forecasts Q. Given $f \in \text{Supp}(Q)$,

$$\frac{1}{T}\sum_{t=1}^{T}\mathbf{1}_{\{f_t=f\}} = \frac{1}{T}\sum_{t=1}^{T}\pi(f \mid p_t) + \frac{1}{T}\sum_{t=1}^{T}\big(\mathbf{1}_{\{f_t=f\}} - \pi(f \mid p_t)\big).$$

The second term goes to zero since it is an average of bounded martingale differences. Moreover,

$$\frac{1}{T}\sum_{t=1}^{T}\pi(f \mid p_t) \to \sum_{p\in D} C_\mu(p)\pi(f \mid p) = Q(f),$$

where the convergence follows from Lemma 2 and the equality holds because π implements Q. Hence, $N_T[f]/T \to Q(f)$ for all $f \in \text{Supp}(Q)$, and therefore $F_{\mu,\tilde{\sigma}} = Q$.

A.3. **Proof of Theorem 1**

We first establish the upper bound. Let I_t indicate whether the sender passes the calibration test in period t. Let

$$d_T := \frac{1}{T}\sum_{t=1}^{T}(1 - I_t)$$

denote the fraction of failure periods up to T.

For each forecast f, let $N_T^+[f]$ and $N_T^-[f]$ denote the number of periods up to T in which f is sent and the test is passed and failed, respectively. Let $\overline{\omega}_T^+[f]$ and $\overline{\omega}_T^-[f]$ denote the corresponding empirical distributions of states. Thus,

$$N_T[f] = N_T^+[f] + N_T^-[f], \qquad d_T = \sum_f \frac{N_T^-[f]}{T}.$$

Consider a horizon T at which the sequence of forecasts up to T is ϵ_T-calibrated. Since

$$\sum_f \frac{N_T[f]}{T}\|\overline{\omega}_T[f] - f\| \leq \epsilon_T,$$

and

$$N_T[f](\overline{\omega}_T[f] - f) = N_T^+[f](\overline{\omega}_T^+[f] - f) + N_T^-[f](\overline{\omega}_T^-[f] - f),$$

the triangle inequality and $\|\overline{\omega}_T^-[f] - f\| \leq \sqrt{2}$ imply

$$\sum_f \frac{N_T^+[f]}{T} \|\overline{\omega}_T^+[f] - f\| \leq \epsilon_T + \sqrt{2} d_T.$$

The sender's average payoff up to T is

$$\frac{1}{T}\sum_{t=1}^{T} u_S(\omega_t, a_t) = \sum_f \frac{N_T^+[f]}{T} u_S(\overline{\omega}_T^+[f], \hat{a}(f)) - c d_T.$$

Since

$$u_S(\overline{\omega}_T^+[f], \hat{a}(f)) \leq \hat{u}_S(f) + L\|\overline{\omega}_T^+[f] - f\|,$$

we obtain

$$\frac{1}{T}\sum_{t=1}^{T} u_S(\omega_t, a_t) \leq \sum_f \frac{N_T^+[f]}{T} \hat{u}_S(f) + L\epsilon_T + (\sqrt{2}L - c) d_T.$$

Since $N_T^+[f] = N_T[f] - N_T^-[f]$ and $|\hat{u}_S(f)| \leq L$,

$$\sum_f \frac{N_T^+[f]}{T} \hat{u}_S(f) \leq \sum_f \frac{N_T[f]}{T} \hat{u}_S(f) + L d_T.$$

Therefore,

$$\frac{1}{T}\sum_{t=1}^{T} u_S(\omega_t, a_t) \leq \sum_f \frac{N_T[f]}{T} \hat{u}_S(f) + L\epsilon_T - \kappa d_T,$$

where

$$\kappa := c - (1 + \sqrt{2})L > 0$$

by Assumption 2.

Along any sequence of such horizons, $\epsilon_T \to 0$. By the same argument as in the proof of Lemma 1, every limit point of the empirical distribution of forecasts belongs to $\mathcal{M}(C_\mu)$. Since $\hat{u}_S$ is upper semicontinuous under the maintained tie-breaking convention, it follows that

$$\limsup_{T\to\infty} \left[\frac{1}{T}\sum_{t=1}^{T} u_S(\omega_t, a_t) + \kappa d_T \right] \leq \mathrm{Per}(C_\mu) \tag{A.9}$$

along horizons at which the sequence of forecasts is ϵ_T-calibrated. Since $\kappa > 0$, this also implies that the sender's average payoff is bounded above by the persuasion value. We keep the stronger inequality because it will also imply that every optimal strategy fails the calibration test in a vanishing fraction of periods in expectation.

We now extend this bound to all horizons. Define

$$Z_T := \sum_{t=1}^{T} u_S(\omega_t, a_t) + \kappa \sum_{t=1}^{T} (1 - I_t).$$

If the sender fails the test in period t, then

$$Z_t - Z_{t-1} = -c + \kappa = -(1 + \sqrt{2})L.$$

Since $\mathrm{Per}(C_\mu) \geq -L$,

$$Z_t - Z_{t-1} \leq \mathrm{Per}(C_\mu).$$

Thus, during failure periods,

$$Z_t - t\,\mathrm{Per}(C_\mu)$$

cannot increase.

If the sender passes the test in period t, then the sequence of forecasts up to $t-1$ is ϵ_{t-1}-calibrated, so (A.9) applies at $t-1$. Since one additional period changes Z_t by only a bounded amount, it follows that

$$\limsup_{T\to\infty} \frac{Z_T}{T} \leq \mathrm{Per}(C_\mu) \qquad \text{almost surely.}$$

If there are only finitely many passing periods, the same conclusion follows immediately since thereafter every period is a failure.

Since Z_T/T is uniformly bounded, the reverse Fatou lemma gives

$$\limsup_{T\to\infty} \mathbb{E}_{\sigma,\mu}\left[\frac{1}{T}\sum_{t=1}^{T} u_S(\omega_t, a_t) + \kappa d_T\right] \leq \mathrm{Per}(C_\mu). \tag{A.10}$$

Since $\kappa d_T \geq 0$, the sender's expected long-run payoff under any forecasting strategy is at most $\mathrm{Per}(C_\mu)$.

We now show that this bound is attained. Let

$$Q^\star \in \arg\max_{Q\in\mathcal{M}(C_\mu)} \mathbb{E}_Q[\hat{u}_S].$$

Since $\mathrm{Supp}(C_\mu)$ is finite, $Q^\star$ can be chosen with finite support. Let $\pi^\star$ be a signaling policy that implements $Q^\star$, and define the forecasting strategy

$$\sigma^\star_t(\cdot \mid p_t) = \pi^\star(\cdot \mid p_t)$$

in every period.

By Lemma 1, this strategy is calibrated, induces $Q^\star$, and fails the calibration test with expected long-run frequency zero. Consider the benchmark in which the receiver plays $\hat{a}(f_t)$ in every period. By the same martingale and ergodic arguments as in Lemma 1, the average benchmark payoff converges almost surely to

$$\sum_p C_\mu(p) \sum_f \pi^\star(f \mid p) \sum_\omega p(\omega) u_S(\omega, \hat{a}(f)) = \sum_f Q^\star(f)\hat{u}_S(f) = \mathrm{Per}(C_\mu).$$

Since payoffs are bounded, the benchmark payoff also converges in expectation to $\mathrm{Per}(C_\mu)$. By Lemma 1, failures occur in a vanishing expected fraction of periods, so they do not affect the expected long-run average payoff. Hence, the strategy attains $\mathrm{Per}(C_\mu)$, which completes the proof.

Finally, (A.10) implies that every optimal forecasting strategy fails the calibration test in a vanishing fraction of periods in expectation. Indeed, for an optimal strategy $\sigma^\star$, the expected average payoff converges to $\mathrm{Per}(C_\mu)$. Since $\kappa > 0$, it follows that $\mathbb{E}_{\sigma^\star,\mu}[d_T] \to 0$.

A.4. **Proposition 5**

Proposition 5. *Fix an arbitrary stochastic process μ. Let σ be a forecasting strategy whose forecasts take values in a finite set $F_0 \subset F$. Then σ is calibrated if and only if*

$$\limsup_{T\to\infty} \sum_{f\in F_0} \frac{N_T[f]}{T} \left\| f - \sum_{p\in D} p \frac{N_T[f,p]}{N_T[f]} \right\| = 0 \quad \mathbb{P}_{\sigma,\mu}\text{-a.s.} \tag{A.11}$$

where $N_T[f,p] = \sum_{t=1}^{T} \mathbf{1}_{\{f_t=f,\, p_t=p\}}$.[24]

Proof. ($\Rightarrow$) First, we show that if the forecasting strategy σ is calibrated, then (A.11) holds. Using the triangle inequality, for any $f \in F_0$,

$$\frac{N_T[f]}{T}\|f - \sum_{p\in D} p\frac{N_T[f,p]}{N_T[f]}\| \le \frac{N_T[f]}{T}\|f - \overline{\omega}_T[f]\| + \|\frac{1}{T}\sum_{t=1}^{T} \mathbf{1}_{\{f_t=f\}}\big(\delta_{\omega_t} - p_t\big)\|.$$

The first term converges to zero since σ is calibrated. For the second term, define

$$X_t := \mathbf{1}_{\{f_t=f\}}\big(\delta_{\omega_t} - p_t\big).$$

Then $(X_t)_{t\ge1}$ is a bounded martingale difference sequence with respect to the natural filtration. Hence, by the strong law of large numbers for bounded martingale differences,

$$\frac{1}{T}\sum_{t=1}^{T} X_t \to 0 \qquad \mathbb{P}_{\sigma,\mu}\text{-a.s.}$$

Since F_0 is finite, we may take the intersection of these probability one events over all $f \in F_0$. Therefore,

$$\limsup_{T\to\infty} \sum_{f\in F_0} \frac{N_T[f]}{T}\|f - \sum_{p\in D} p\frac{N_T[f,p]}{N_T[f]}\| = 0 \quad \mathbb{P}_{\sigma,\mu}\text{-a.s.}$$

($\Leftarrow$) Conversely, suppose that (A.11) holds. Fix $f \in F_0$. Using the triangle inequality,

$$\frac{N_T[f]}{T}\|f - \overline{\omega}_T[f]\| \le \frac{N_T[f]}{T}\|f - \sum_{p} p\frac{N_T[f,p]}{N_T[f]}\| + \|\frac{1}{T}\sum_{t=1}^{T} \mathbf{1}_{\{f_t=f\}}\big(\delta_{\omega_t} - p_t\big)\|.$$

The first term converges to zero by assumption, and the second converges to zero by the same martingale argument as above. Since F_0 is finite, summing over $f \in F_0$ yields

$$\limsup_{T\to\infty} \sum_{f\in F_0} \frac{N_T[f]}{T}\|f - \overline{\omega}_T[f]\| = 0 \quad \mathbb{P}_{\sigma,\mu}\text{-a.s.}$$

□

A.5. **Lemma 2**

A stochastic process $(\omega_t)_{t\ge1}$ is *stationary* if, for any $k \in \mathbb{N}$, the joint distribution of the k-tuple $(\omega_t, \omega_{t+1}, \ldots, \omega_{t+k-1})$ does not depend on t. Let $T : \Omega^\infty \to \Omega^\infty$ be the shift transformation given by $T(\omega)_t = \omega_{t+1}$ for all $t \in \mathbb{N}$. Let $\mathcal{I}$ denote the $\sigma-$algebra of all

[24] If $N_T[f] = 0$ in (A.11), we set the entire summand equal to zero.

invariant Borel sets for the transformation T. The stationary process $(\omega_t)_{t\geq 1}$ is *ergodic* if $\mathcal{I}$ is trivial, that is, $\mathbb{P}(A) \in \{0,1\}$ for all $A \in \mathcal{I}$.

Lemma 2. *For a stationary ergodic process μ, the distribution of conditionals C_μ exists and is constant almost surely.*

Proof. Consider the two-sided extension of the stationary process so that $\mu \in \Delta(\Omega^{\mathbb{Z}})$. Let $\omega_a^b = (\omega_a, \ldots, \omega_{b-1})$ where $a < b$. Define

$$f_n := \mu(\omega_0 = \cdot \mid \omega_{-n}^0), \qquad f_\infty := \mu(\omega_0 = \cdot \mid \omega_{-\infty}^0).$$

By the martingale convergence theorem, $f_n \to f_\infty$ almost surely. By stationarity,

$$p_n := \mu(\omega_n = \cdot \mid \omega_0^n) = f_n \circ T^n.$$

where T is the shift transformation. Recall that p_n takes values in the finite set $D \subset \Delta(\Omega)$; since μ is T-invariant and $p_n = f_n \circ T^n$, it follows that f_n also takes values in D.

Fix $p \in D$ and let $g_n = \mathbf{1}_{\{f_n = p\}}$ and $g_\infty = \mathbf{1}_{\{f_\infty = p\}}$. Since $f_n \to f_\infty$ almost surely and D is finite, we have $g_n \to g_\infty$ almost surely, and $\sup_n |g_n| \leq 1 \in L_1(\mu)$. Maker's ergodic theorem (see Kallenberg, 2002) implies

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} g_n \circ T^n = \mathbb{E}[g_\infty \mid \mathcal{I}] \qquad \mathbb{P}_\mu\text{-a.s.},$$

where $\mathcal{I}$ is the shift-invariant σ-algebra. Since $g_n \circ T^n = \mathbf{1}_{\{p_n = p\}}$, this becomes

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} \mathbf{1}_{\{p_n = p\}} = \mathbb{E}[\mathbf{1}_{\{f_\infty = p\}} \mid \mathcal{I}] \qquad \mathbb{P}_\mu\text{-a.s.}$$

If μ is ergodic then $\mathcal{I}$ is trivial, so the right-hand side equals $\mathbb{E}[\mathbf{1}_{\{f_\infty = p\}}]$, which is constant. Hence, for each $p \in D$, the limit

$$C_\mu(p) := \lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} \mathbf{1}_{\{p_n = p\}}$$

exists and is constant almost surely. □

A.6. **Proof of Theorem 2**

The main idea of the proof is to combine the sender's payoff and calibration cost to form a vector-valued payoff. We then identify a suitable closed and convex target set that captures the definition of attainable function (a similar approach appears in Mannor et al., 2009). Finally, we apply the dual condition of approachability to demonstrate that the set is approachable, ensuring convergence regardless of nature's strategy.

Given a forecast $f \in F_\epsilon$ and state $\omega \in \Omega$, the calibration cost vector is given by

$$c(f, \omega) = (\mathbf{0}, \ldots, f - \delta_\omega, ..., \mathbf{0}) \in \mathbb{R}^{|F_\epsilon||\Omega|}. \tag{A.12}$$

It is a vector of $|F_\epsilon|$ elements of size $\mathbb{R}^{|\Omega|}$ with one non-zero element (at the position for f) while the rest are equal to $\mathbf{0} \in \mathbb{R}^{|\Omega|}$.

In each period t, the sender and nature simultaneously choose $f_t \in F_\epsilon$ and $\omega_t \in \Omega$ respectively, resulting in payoff $r_t = u_S(\omega_t, \hat{a}(f_t))$ and calibration cost $c_t = c(f_t, \omega_t)$. The ϵ-calibration condition (2.3) can be rewritten as follows: the average of the sequence of vector-valued calibration costs $c_t = c(f_t, \omega_t)$ converges almost surely to the set $\mathcal{C}$, where[25]

$$\mathcal{C} = \{c \in \mathbb{R}^{|F_\epsilon||\Omega|} : \sum_{f \in F_\epsilon} \|\underline{c}_f\| \leq \epsilon\}.$$

We now show the function $p \mapsto \overline{\text{Co}}\, \hat{u}_S^\epsilon$ is attainable. For each period t, consider the vector-valued payoff

$$m(f_t, \omega_t) := (r_t, c_t, \delta_{\omega_t}) \in \mathbb{R} \times \mathbb{R}^{|F_\epsilon||\Omega|} \times \Delta(\Omega). \tag{A.13}$$

Define the sets:

$$\mathcal{T}_1 = \{(r, c, p) : r \geq \overline{\text{Co}}\, \hat{u}_S^\epsilon(p)\}, \quad \mathcal{T}_2 = \{(r, c, p) : c \in \mathcal{C}\},$$

and let $\mathcal{T} = \mathcal{T}_1 \cap \mathcal{T}_2$ denote the target set. The set $\mathcal{T}$ is closed and convex.

To show that the average of $m(f_t, \omega_t)$ approaches $\mathcal{T}$, it suffices to verify the dual condition of approachability [Blackwell, 1956], namely that for every $p \in \Delta(\Omega)$ there exists $x \in \Delta(F_\epsilon)$ such that $m(x, p) \in \mathcal{T}$, where

$$m(x, p) = \Big(\sum_{f \in F_\epsilon} x(f) \sum_{\omega \in \Omega} p(\omega) u_S(\omega, \hat{a}(f)),\ \sum_{f \in F_\epsilon} \sum_{\omega \in \Omega} x(f) p(\omega) c(f, \omega),\ p \Big). \tag{A.14}$$

Fix $p \in \Delta(\Omega)$ and choose $x = \delta_{f^*(p)} \in \Delta(F_\epsilon)$, that is, the degenerate distribution that assigns probability one to the forecast $f^*(p) \in F_\epsilon$. We verify that this choice satisfies the dual condition.

With this choice, the sender's payoff equals $\hat{u}_S^\epsilon(p)$. Hence, by the definition of closed convex hull, it follows that $m(x, p) \in \mathcal{T}_1$. Moreover, $\sum_\omega p(\omega) c(f^*(p), \omega) = (\mathbf{0}, \ldots, f^*(p) - p, \ldots, \mathbf{0})$, so the calibration error equals $\|f^*(p) - p\| \leq \epsilon$, and hence $m(x, p) \in \mathcal{T}_2$. Therefore $m(x, p) \in \mathcal{T}$, the dual condition holds, and the set $\mathcal{T}$ is approachable.

Finally, we show that approachability of $\mathcal{T}$ implies the two conditions of attainability. Since F_ϵ is finite, $\hat{u}_S^\epsilon$ is piecewise affine, and its closed convex envelope is continuous. Thus, approaching $\mathcal{T}$ ensures that the sender's average payoff exceeds $\overline{\text{Co}}\, \hat{u}_S^\epsilon(\overline{\omega}_T)$ while remaining ϵ-calibrated.

A.7. **Proof of Proposition 1**

We use the notion of opportunistic approachability, introduced in Bernstein et al. [2014]. The strategies proposed there not only ensure approachability of the target set in the adversarial case, but also exploit restrictions in nature's play to approach smaller subsets when such restrictions arise, either in a statistical or an empirical sense; we focus on the latter.

[25] Formally, this implies $\lim_{T \to \infty} \text{dist}(\frac{1}{T} \sum_{t=1}^T c(f_t, \omega_t), \mathcal{C}) = 0$ almost surely, where dist(.) is the Euclidean distance.

Definition 8 (Bernstein et al. [2014])**.** The play of nature is *empirically Q-restricted* with respect to a partition $\{l_m\}$, if there exists a convex subset $Q \subseteq \Delta(\Omega)$ such that,

$$\lim_{M\to\infty} \frac{1}{n_M} \sum_{m=1}^{M} l_m \operatorname{dist}(\overline{\omega}_m, Q) = 0, \quad \mathbb{P}_\mu\text{-a.s.}$$

where l_m denotes the length of block m, $n_M = \sum_{m=1}^{M} l_m$ denotes the time at the end of block M, and $\overline{\omega}_m$ denotes the empirical distribution of states in block m.

Apply the opportunistic approachability strategy to the vector payoff m used in the proof of Theorem 2. If nature's play is empirically Q-restricted with respect to a suitable polynomially growing partition, then Theorem 5 of Bernstein et al. [2014] implies that the average vector payoff approaches the corresponding closed convex image. In particular, the sender's average payoff approaches

$$R^+(Q) := \bigcap_{\eta>0} \operatorname{Co}\Big\{\hat{u}_S^\epsilon(q) : d(q, Q) \le \eta\Big\}.$$

In words, $R^+(Q)$ is the closed convex image of the approximate indirect utility restricted to Q. The fully adversarial case corresponds to $Q = \Delta(\Omega)$.

By assumption, the order-k Markov chain is such that the augmented chain on Ω^k is irreducible and aperiodic, and therefore admits a unique invariant distribution. Let $p \in \Delta(\Omega)$ denote its invariant marginal distribution over states.

Fix $\eta > 0$. Concentration inequalities for Markov chains imply that there exist constants $C_\eta, c_\eta > 0$ such that, for every block length l,

$$\mathbb{P}\big(\|\overline{\omega}_l - p\| > \eta\big) \le C_\eta e^{-c_\eta l},$$

see, e.g., Lezaud [1998] or Paulin [2015].

Choose a partition into consecutive blocks with lengths $l_m = \lceil m^\nu \rceil$ for some $\nu \in (0, 1)$. By the Borel–Cantelli lemma, for every $\eta > 0$, almost surely only finitely many blocks satisfy $\|\overline{\omega}_m - p\| > \eta$. Hence, $\overline{\omega}_m \to p$ almost surely, and nature's play is empirically $\{p\}$-restricted with respect to the partition $(l_m)_{m\ge1}$. For sufficiently small ν, these block lengths satisfy the growth condition of Theorem 5 of Bernstein et al. [2014]. Consequently, the distance between the sender's average payoff and $R^+(\{p\})$ converges to zero.

If p lies in the interior of a grid cell, then $R^+(\{p\}) = \{\hat{u}_S^\epsilon(p)\}$. If p lies on the boundary of two or more grid cells, choose $f^*(p)$ among the associated forecasts to minimize the sender's expected payoff at p. Since f^* is constant within each adjacent cell, every limit of $\hat{u}_S^\epsilon(q)$ as $q \to p$ is at least $\hat{u}_S^\epsilon(p)$. Hence, the sender attains $\hat{u}_S^\epsilon(p)$. The same strategy is also ϵ-calibrated, since its average calibration cost approaches $\mathcal{C}$, as in the proof of Theorem 2.

A.8. **Proof of Proposition 2**

Fix any calibrated forecasting strategy and suppose the receiver best responds to every forecast, so that $a_t = \hat{a}(f_t)$ for all t. Thus, whenever $N_T[f] > 0$, we have $\overline{u}_{R,T}[f] = u_R(\overline{\omega}_T[f], \hat{a}(f))$.

Let $L_R := \max_{a \in A} \|u_R(\cdot, a)\|$. Since $\hat{a}(f)$ is a best response to f, for every $a \in A$,

$$u_R(\overline{\omega}_T[f], a) - u_R(\overline{\omega}_T[f], \hat{a}(f)) \leq 2L_R \|\overline{\omega}_T[f] - f\|.$$

Therefore,

$$\begin{aligned} &\limsup_{T\to\infty} \frac{N_T[f]}{T} \left(\max_{a \in A} u_R(\overline{\omega}_T[f], a) - u_R(\overline{\omega}_T[f], \hat{a}(f)) \right) \\ &\leq 2L_R \limsup_{T\to\infty} \frac{N_T[f]}{T} \|\overline{\omega}_T[f] - f\| = 0, \end{aligned}$$

where the equality follows from calibration. Hence, the receiver has no regret with respect to f.

A.9. **Proof of Proposition 3**

Let $Q^\star \in \mathcal{M}(C_\mu)$ be an optimal distribution with finite support, and let $\pi^\star$ be a signaling policy that implements it.

By the nondegeneracy condition, for each $f \in \text{Supp}(Q^\star)$ there exists a belief $q_f \in \text{Co}(\text{Supp}(C_\mu))$ at which $\hat{a}(f)$ is the unique best response.

We construct an auxiliary signaling policy ρ that assigns each $f \in \text{Supp}(Q^\star)$ positive probability β_f and induces posterior mean q_f. Write $q_f = \sum_{p \in \text{Supp}(C_\mu)} \lambda_f(p) p$, where $\lambda_f \in \Delta(\text{Supp}(C_\mu))$. Choose $\beta_f > 0$ sufficiently small that

$$\sum_{f \in \text{Supp}(Q^\star)} \beta_f \lambda_f(p) \leq C_\mu(p) \qquad \text{for every } p \in \text{Supp}(C_\mu).$$

For each such f and p, set $\rho(f \mid p) = \beta_f \lambda_f(p) / C_\mu(p)$ and assign the remaining probability to a forecast $g \notin \text{Supp}(Q^\star)$.

Fix $\eta \in (0, 1)$ and let the sender use

$$\pi^\eta = (1 - \eta)\pi^\star + \eta\rho$$

in every period. By Lemma 2 and the strong law for bounded martingale differences, for every $f \in \text{Supp}(Q^\star)$ and $a \neq \hat{a}(f)$, almost surely,

$$\begin{aligned} \frac{1}{t}\left(U^f_{\hat{a}(f),t} - U^f_{a,t}\right) \to\ & (1 - \eta) Q^\star(f) \big(u_R(f, \hat{a}(f)) - u_R(f, a)\big) \\ & + \eta \beta_f \big(u_R(q_f, \hat{a}(f)) - u_R(q_f, a)\big) > 0. \end{aligned}$$

The first term is nonnegative because $\hat{a}(f)$ is a best response to f, and the second is strictly positive by construction.

Since $\text{Supp}(Q^\star)$ and A are finite, there exists $d_\eta > 0$ such that all these cumulative payoff differences exceed $d_\eta t$ for all sufficiently large t, almost surely. In the T-period

game, the mean-based threshold is $\gamma_T T$, where $\gamma_T = o(1)$. Thus, the fraction of periods in which these differences fail to exceed $\gamma_T T$ converges to zero almost surely.

For any fixed mean-based receiver, bounded convergence and the mean-based property imply that the receiver chooses $\hat{a}(f)$ in all but $o(T)$ periods in expectation following forecasts in $\text{Supp}(Q^\star)$.

Under π^η, the sender uses $\pi^\star$ with probability $1-\eta$. By stationarity and ergodicity, these periods would contribute $(1-\eta)\text{Per}(C_\mu)$ to the limiting expected average payoff if the receiver always chose $\hat{a}(f)$. Since payoffs are bounded, the vanishing frequency of other actions does not affect this limit. The auxiliary rule is used with probability η, and its payoff is bounded below by $-L$. Therefore,

$$\liminf_{T\to\infty} \frac{1}{T}\mathbb{E}\left[\sum_{t=1}^{T} u_S(\omega_t, a_t)\right] \geq (1-\eta)\text{Per}(C_\mu) - \eta L \geq \text{Per}(C_\mu) - 2L\eta.$$

Finally, let $\eta_n = 1/n$ for $n \geq 2$. For each n, choose T_n sufficiently large that using π^{η_n} throughout the T-period game guarantees an expected average payoff of at least $\text{Per}(C_\mu) - (2L+1)/n$ whenever $T \geq T_n$, with T_n strictly increasing. For each horizon $T_n \leq T < T_{n+1}$, the sender uses π^{η_n} in every period. As $T \to \infty$, the chosen η_n converges to zero, so the sender guarantees $\text{Per}(C_\mu)$.